

DSA Day 19: Trees - terminology & representation

Book: Data Structures Through C in Depth - Ch6 §6.1-6.8 · pp 176-185 | Code in Python

Overview

Trees are hierarchical. A binary tree has ≤ 2 children. Store as linked nodes or, for complete trees, in an array (children of i at $2i+1$, $2i+2$).

Key concepts to master

- root/leaf/height/level/degree
- strict/full/complete binary tree
- Array vs linked representation

Python implementation

```
class TNode:
    def __init__(self,d): self.data=d; self.left=None; self.right=None
# array form: children of i are 2i+1 and 2i+2
```

Complexity

Height h ; a balanced tree has $h \sim \log n$.

Common pitfalls

- Confusing height vs depth
- Array form wastes space for sparse trees

Practice

- Build a 7-node tree
- Find children of index i in the array form

